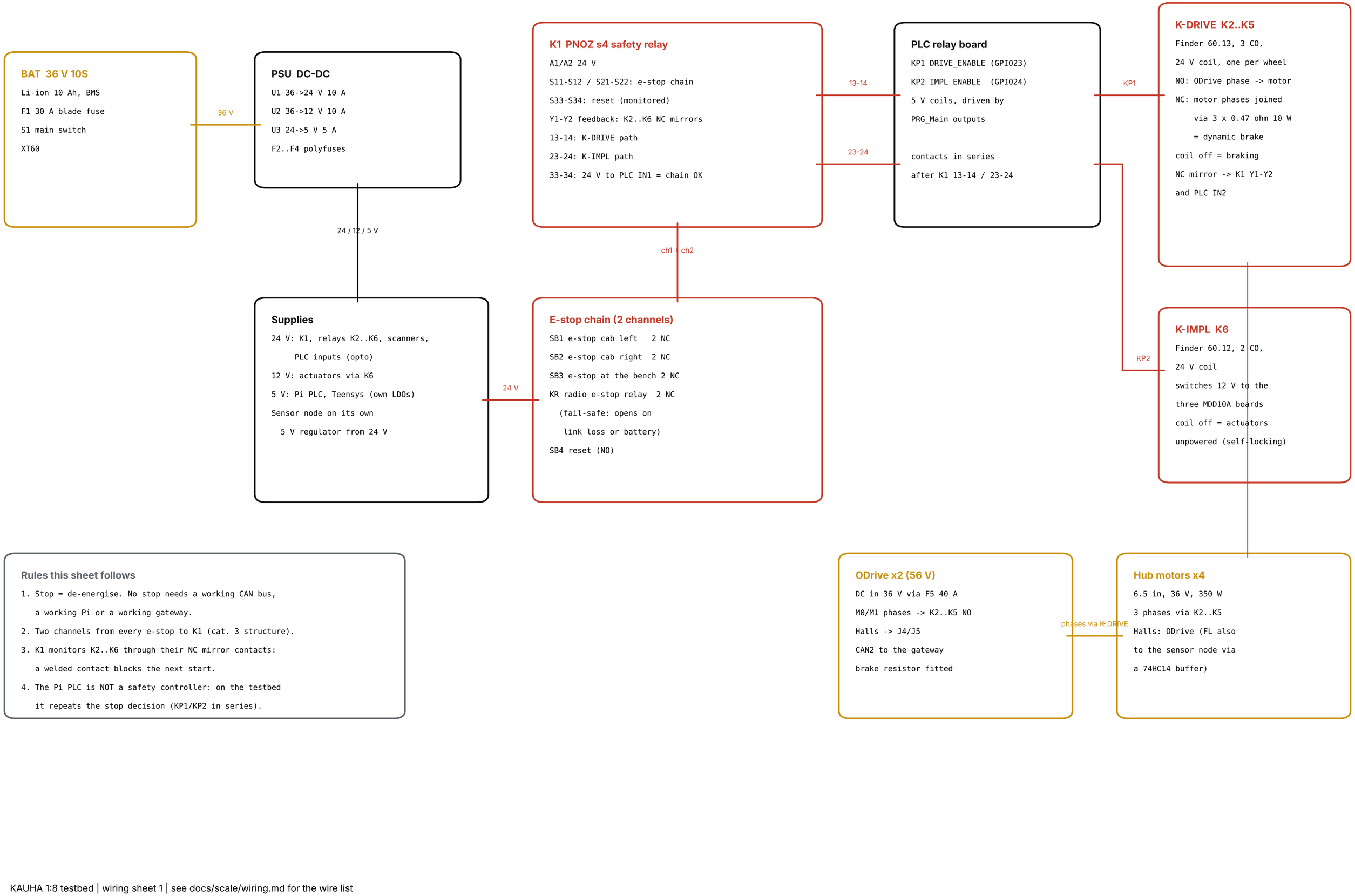
KAUHA 1:8 testbed - sheet 1: power, stop chain and hardwired outputs

Red: safety chain (24 V). Amber: 36 V traction. Black: logic. Every stop path de-energises: a broken wire, a dropped relay or a dead PLC all stop the machine.



Rules this sheet follows

1. Stop = de-energise. No stop needs a working CAN bus,
   a working Pi or a working gateway.
2. Two channels from every e-stop to K1 (cat. 3 structure).
3. K1 monitors K2..K6 through their NC mirror contacts:
   a welded contact blocks the next start.
4. The Pi PLC is NOT a safety controller: on the testbed
   it repeats the stop decision (KP1/KP2 in series).

KAUHA 1:8 testbed - sheet 2: CAN bus and signals

Machine bus: 500 kbit/s, 29-bit J1939 identifiers, hardware/can/kauha_machine.dbc. Twisted pair, 120 ohm at both ends only. Motor bus (CAN2) is separate.

Frame timing and timeouts (PLC side)

```
RemoteHeartbeat 20 Hz   timeout 0.5 s  -> REMOTE_LINK
DriveCmd        20 Hz   timeout 0.2 s  -> WATCHDOG
MachineState1   40 Hz   stale 0.1 s    -> SPEED_DISCREPANCY
SafeSensors     40 Hz   stale 0.1 s    -> SCANNER_FAULT
ScannerStatus   40 Hz   stale 0.1 s    -> SCANNER_FAULT
ZoneStatus      10 Hz   stale 0.5 s    -> POSITION_INVALID (2 s)
```

Every safety-relevant frame

byte 6 low nibble: alive counter (+1 per frame)

byte 7: CRC-8 SAE J1850 over data ID + bytes 0..6

data ID = PGN low byte (unique per message)

checked in FB_E2ERx (PLC) and kauha_e2e.c (firmware)

both tested bit for bit against kauha/can/e2e.py

Connectors

CAN: 4-pin M8 or JST-XH: CAN_H, CAN_L, GND, shield

Stub length < 0.3 m at 500 kbit/s

E-stop and relay wiring: 0.5 mm2, ferrules,

  two colours for the two channels

Traction: 2.5 mm2 silicone, XT60/XT90

CAN_H / CAN_L (machine bus)

120R

120R

Safety PLC 0x81

Pi 4 + CAN HAT (can0)

CODESYS PRG_Main 25 ms

tx SafetyStatus 40 Hz

rx all frames below

GPIO: IN1..IN8, OUT1..OUT9

Gateway 0x83

Teensy 4.1 CAN1

rx DriveCmd, ImplementCmd,

   SafetyStatus

tx MachineState1/2

CAN2 -> ODrives (11-bit)

PWM -> 3 x MDD10A

Sensor node 0x84

Teensy 4.0

AS5600 hinge (Wire)

AS5600 boom (Wire1)

FL Halls via 74HC14

tx SafeSensors 40 Hz

Field evaluator 0x85

evaluator profile only:

Pi + 2 x LD19 lidar

kauha/scale/field_eval.py

tx ScannerStatus 40 Hz

(rated scanners: wired to

 PLC IN3..IN8 instead)

Autonomy 0x80

Pi 5 / Jetson, ROS 2

kauha_ros can_bridge

tx DriveCmd, ImplementCmd

rx SafetyStatus, MachineState

USB-CAN or CAN HAT

Remote 0x82 / zones 0x86

supervisor laptop:

tx RemoteHeartbeat 20 Hz

tx ZoneStatus 10 Hz

(console + test bench

 zone switches)